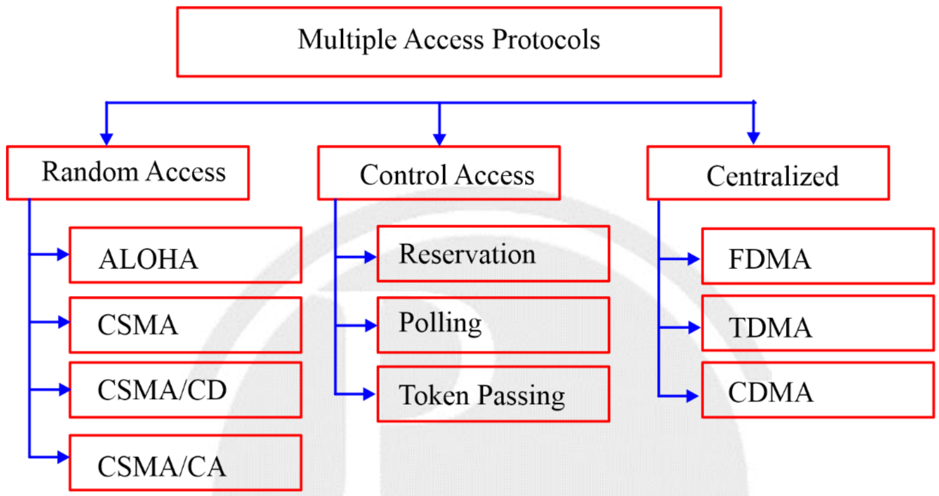
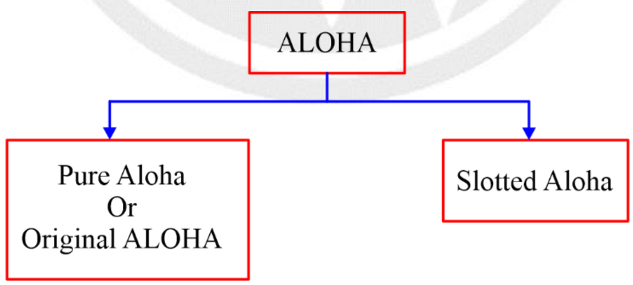
6 MEDIUM ACESS CONTROL[MAC]



6.1 Aloha

Aloha was developed at university of Hawaii in 1970’s.
It was designed for wireless LAN but it can be used in any shared medium.
Each station sends equal size frame.



Pure Aloha	Slotted Aloha
Any station transmits the data at any time.	Any station can transmit the data at the beginning of any time slot.
Vulnerable time in which collision may occur = 2 * T_f (T_f - Transmission time for single frame)	Vulnerable time in which collision may occur = T_f
Throughput of pure aloha = $G * e^{-2G}$	Throughput of slotted Aloha = $G * e^{-G}$
Maximum throughput s_{max} = 18.4 % (When G = 1/2)	Maximum throughput s_{max} = 36.8 % (When G = 1)
The main advantage of pure aloha is it simplicity in implementation	The main advantage of slotted aloha is that it reduces the number of collisions to Half and double the throughput of pure aloha

6.2 CSMA (Carrier Sense multiple access)

CSMA requires that each station, first sense the carrier before transmitting the data.
Vulnerable time for CSMA = Propagation time.

6.3 Persistence methods in CSSMA

Persistent
Non-persistent
P-persistent

6.4 CSMA/CD (Carrier Sense multiple access/Collision Detection)

1. Minimum size of frame to detect the collision in Ethernet (CSMA/CD)
   $T_d \geq 2 * P_d + T_d$(Jam signal)
2. Backoff Algorithm
   Waiting time = K * Slot duration
   = K * RTT
   = $K * 2 * P_d$

   K is any random number in between 0 to 2^n -1.
   n is collision number (Collision number is respect to data packet).

3. Efficiency in Ethernet (CSMA/CD)
   $\boxed{\eta = \frac{1}{1+6.44a}}$ or $\eta = \frac{\text{useful time}}{\text{Total time}} = \frac{T_d}{\text{Collision time}+T_d+P_d}$

4. $P(1-P)^{N-1} \rightarrow$ Probability of success for single station(Throughput of Host)

$NP(1-P)^{N-1} \rightarrow$ Probability of success for any station among all stations [Throughput of channel]

5. Ethernet Frame Structure:

Preamble	SFD	DA	SA	Length Of Data	Data	CRC
7B	1B	6B	6B	2B	(46B – 1500B)	4B

PL DLL DLL

Ethernet uses Manchester encoding technique for converting data bits into signal.
(Baud rate = 2 ∗ bit rate)
Bit rate = 1/2 baud rate